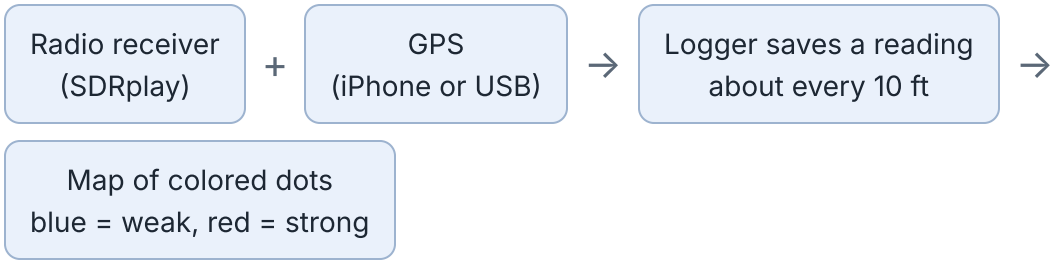
RF Noise Map

User guide — find where radio noise (power lines, poles, noisy buildings) is strongest, by driving a route and drawing the readings as colored dots on a street map.



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1. What you need

- **A computer in the car:** a Mac or a Windows 11 PC (laptop). It runs the program and does the logging.
- **An SDRplay receiver** plugged into that computer, with an antenna mounted away from the engine.
- **A GPS**, either:
  - your **iPhone** with the **GPS2IP** app (simple; one position per second), or
  - a **USB GPS** plugged into the computer. A 10-updates-per-second model gives much closer dot spacing at 40–45 mph.
- The iPhone's **Personal Hotspot** is handy for internet in the car (needed only to draw the street background when you view a map, and for iPhone GPS).

Why not the iPhone as the whole system? The SDRplay needs a computer driver; it cannot run on an iPhone or iPad. The phone is only the GPS and the hotspot.

2. Download and install

Download (Windows and Mac) — no GitHub account needed

1. Open **github.com/kw4ex/RFNoise-Mapping** in your web browser and click **Releases** (on the right side of the page). Click the newest entry, the one marked **Latest**. (A **Pre-release** label means it is still being tested.)
2. Scroll down to **Assets**, at the bottom of that page.
3. **Windows:** click `Install_RFNoise_Windows.bat`. **Mac:** click `RFNoise_Mac_….zip`. Your browser saves it in your **Downloads** folder. If it warns that the file may be harmful, choose **Keep**.
4. **Do not** click **Source code (zip)** or **Source code (tar.gz)**. Those are for programmers.

Windows: you may also click `RFNoise_User_Guide.pdf` in Assets, the printable copy of this guide.

Mac (experimental: not yet tested on a real Mac by the author)

1. Open **Downloads** and double-click the zip you downloaded to unzip it. Keep the folders (**mac** and **app**) together.
2. Open the **mac** folder. **Right-click** `Install_RFNoise.command` and choose **Open**, then **Open** again. (Only the first time. macOS asks because the file was downloaded.)
3. A window explains each step before it does it. Press `y` and Return to go on.
4. When it says **DONE**, an **RF Noise Map** icon is on your Desktop and in **Applications**.

If macOS says "cannot be opened because Apple cannot check it for malicious software" (normal for a downloaded file): click **Done** (never "Move to Trash"). Then either (a) right-click the file, choose **Open**, then **Open** again; or (b) open **System Settings** → **Privacy & Security**, scroll down to the message about the file, click **Open Anyway** and enter your Mac password. You only need to do this for the installer; it then clears the warning for the other files in the folder (including the uninstaller).

If the file opens in a text editor, or says it cannot run: open **Terminal**, type `bash ` (with a space), drag the install file into the Terminal window, press Return.

Windows 11

1. Double-click `Install_RFNoise_Windows.bat` (the file you downloaded; keep it anywhere, for example your Downloads folder). If Windows says **"Windows protected your PC"**, click **More info** then **Run anyway**.
2. A window asks **which folder** to use. Press Enter to accept **C:\RadioScripts**, or type another. The program makes its own subfolder, **RFNoise**, inside it, so everything lives in **C:\RadioScripts\RFNoise**.
3. It explains each step and asks short questions (answer `y` or `n`). It finds Python and offers to install it if missing, and no administrator password is needed.
4. When it says **All done**, start the program with `Start_RFNoise.bat` in that folder, or the Desktop shortcut if you chose one.

The installer keeps a **log of every install** (the newest 10) in the **Logs** folder inside the RFNoise folder, plus a copy named install_log.txt. If something goes wrong it also tells you in plain words what the cause probably is, what to do and where to read more, and offers to make a **support file** (one zip) to send to Claude. While the program runs, problems are written to **Logs/program_errors.log**. In the panel, **Tools › Show recent problems** shows them and **Make a support file** packs everything. The installer also checks your computer first, and if the program is already installed it offers upgrade/repair, a check, or uninstall.

3. The radio driver (SDRplay)

The SDRplay driver is not mine to include; it comes from sdrplay.com. Without it, **Practice mode and all map features still work**, and you can make maps from drives recorded on another computer.

Mac

1. Run the installer again and answer `y` at **STEP 3**. It opens the SDRplay page; download the **macOS API** installer and run it. **Restart the Mac.**
2. Run the installer once more and answer `y` at STEP 3. It installs the helper tools (it will offer to install Homebrew, a standard Mac tool installer; it asks for your Mac password) and builds the SDRplay driver module.
3. Newer driver modules need SDRplay API **3.15 or later**. If the build fails, update the API from sdrplay.com and try again.

Windows

1. **SDRplay driver:** run the installer again and answer `y` at the radio step (or go to sdrplay.com → Downloads → Windows) and install the **API**. Restart if it asks.
2. **Radio library:** the program also needs **PothosSDR**, a free Windows bundle (download link on **github.com/pothosware/PothosSDR/wiki**). Install it with the default choices.
3. Open RF Noise Map, plug in the SDRplay, and press Check setup.
4. **Python version:** PothosSDR's Python part is made for ONE Python version (currently **Python 3.9**). The installer notices this, uses (or installs) the matching Python by itself, and sets up the launcher files to use it. If you install PothosSDR later, run the installer again so it picks the right Python.
5. Close other SDR programs (SDRuno, SDRconnect, SDR Console, HDSDR ...) before you start: only one program can use the radio at a time. RF Noise Map shows a yellow box with a Close them for me button if it sees one.
6. For a complete automatic check, double-click **Test_Everything.bat** (or press Full test in the window). It takes about a minute and tests the radio, the GPS and a pretend drive.

> Press Check setup any time. It says in plain English what is working and what is missing.

4. First try (no hardware)

1. Open **RF Noise Map** (double-click the icon). A window opens in your web browser.
2. Press Try a practice drive + map in the **Tools** box. A fake drive is recorded in a second and a map opens: a red stretch is a pretend noisy power pole.
3. Or tick **Practice mode** in box 1 and press **Start logging** to watch the live readings for about 40 seconds.

The practice route is made up, so it won't follow real streets. Real drives will.

5. Connect the GPS

Option A: iPhone

1. On the iPhone, install **GPS2IP** from the App Store. A free **GPS2IP Lite** lets you try it; the full version (a few dollars) is the one to use.
2. Turn on **Personal Hotspot** on the iPhone (Settings → Personal Hotspot). Join the computer to that Wi-Fi.

3. Open GPS2IP and switch it **on**. Use the fastest update rate (1 second). Make sure the **Sockets** option is on. Leave the app open.
4. The app shows an **address** and a **port**. In RF Noise Map choose **iPhone** and type them in. (`172.20.10.1` and `11123` are the usual values.)
5. Press Check setup, tick **Also listen to the GPS**, and look for a green ✓ with your position.

> The iPhone updates once per second. At 45 mph that is about 66 ft between true GPS fixes, so the program slides the position forward along your direction of travel. This works well on straight roads and is less exact in sharp turns.

Option B: USB GPS

1. Plug it into the computer. In RF Noise Map choose **USB GPS** and press Find my GPS. It listens on every likely COM port at the common speeds (4800, 9600, 38400, 115200) and fills in the right **Port** and **Speed** for you. (Or double-click **Find_GPS.bat**.) Many pucks use 4800, not 9600.
2. If it says the port is in use, another program has it open: close other GPS or mapping programs, or unplug the puck for 5 seconds and plug it back in. A new puck often needs a few minutes outdoors for its first position fix.
3. Press Check setup, tick **Also listen to the GPS**, and look for a green ✓.
4. Most pucks come set to 1 update per second. For about 10 ft spacing at 45 mph you want one set to **10 updates per second** (a one-time setting). Ask Claude and we'll do it together.

6. Collect data

1. In box **1 Collect data**, type a **Route name** (optional, but it makes drives easy to find).
2. Choose the GPS (iPhone or USB), unless you're in Practice mode.
3. Enter up to **four frequencies** in MHz. Suggestions: **0.75** (AM broadcast band, wide area), **3.7** or **7.3** (ham bands), and a VHF channel such as **148** or an **AM airband channel (118–137)** for pinpointing.
   - **Pick a quiet channel** with no station on it. A pilot's transmission on an airband channel would look like a hot spot.
   - **Width** is the slice measured (5–10 kHz is good). Narrow slices ignore nearby stations.
   - Leave **Calibration** at 0 unless you've calibrated (section 10).
4. **Dot spacing** sets how often a reading is saved (default one every 10 ft). Use the quick buttons or type a number. You can change it **while driving**; it applies from the next dot. Smaller spacing gives more detail but bigger files, and if you drive faster than the radio can measure, dots end up farther apart than the setting.
5. **Take a reading by time instead of distance.** Next to Dot spacing, choose **by time** and type the seconds (for example 5). The program then saves a reading every few seconds, even when you stand still — handy for catching an **intermittent noise** at one spot. The default is by distance. You can switch while driving; readings taken at one place pile up on one dot on the map.
6. Press ▶ Start logging. The header shows **Logging** and **GPS: OK**, and the big numbers show the live level for each frequency.
7. **Drive the route.** Steady speed is best. You can change frequencies **while driving**: edit the boxes and press Apply frequency changes now. Pause / resume is there for stops.
8. Press ■ Stop logging when finished. If **Make a map automatically** is ticked, the map opens.

Good habits: don't change attenuation mid-route (dots before and after would not be directly comparable); keep the computer plugged in (the program keeps it awake while logging); drive each important road twice, once in each direction; note the date and conditions in **Name / notes**.

Safety: have a passenger operate the computer, or set everything up while parked. Please don't use the screen while driving.

Panadapter: see and hear the band, then pick your frequencies

The **Panadapter** box (just below Collect data) is a live picture of a slice of the radio spectrum, like the display on a ham SDR. The top picture shows signal strength for every frequency in view; the coloured picture below it is the **waterfall**, which scrolls so you can see signals come and go over time.

1. Press ▶ Start panadapter. Allow sound if the browser asks. (It cannot run while logging, because the radio can only do one job at a time. Starting a log closes it automatically.)
2. Pick a band in **Jump to band** (this includes CB). Tick **Widen the view to show the whole band** first if you want the Width of view set automatically so the whole band fits (bands wider than 8 MHz, such as 70 cm, show their middle part). Or type a **Centre** frequency in MHz. **Width of view** is how much spectrum you see (250 kHz to 8 MHz). The grey edges of the picture are less accurate. **Drag the picture left or right** (or swipe sideways with two fingers on a trackpad) to slide the view to other frequencies. If you type a frequency in **Tuned** that is off the screen, the view re-centres on it.
3. **Click a signal** to tune to it and hear it. The red line is the tuned frequency and the red band is the receiver filter. The mouse wheel, or the Up and Down keys in the **Tuned** box, change frequency one **Step** at a time (set the step beside it, from 10 Hz to 25 kHz), or type a frequency in **Tuned**.
4. Choose the **Mode** (AM, **SAM**, USB, LSB, CW, narrow FM or **broadcast FM**). SAM is "synchronous AM": it locks onto the station's carrier, so a station that sounds distorted or warbly in plain AM (from fading or slight mistuning) often sounds cleaner. If it is still harsh, the radio may be overloaded — raise the IF attenuation. Broadcast FM (88–108 MHz, mono) has its own band choice, **FM broadcast**. **Left and Right arrow keys** tune down and up by the chosen step (hold Shift for ten steps). Under the picture, **Reset to defaults** puts IF attenuation back to 40 dB and RF attenuation to 0. Also choose the **Bandwidth** of the filter, as on any ham radio. The small green picture shows the audio you are hearing. Use **Squelch** to silence the audio when nothing is there.
5. **Known-frequency markers:** small yellow labels on the picture mark the 40 CB channels, time stations (WWV, CHU), NOAA weather, ham calling and beacon frequencies, airband and marine channels and FRS/GMRS. Click a label to tune it with a suitable mode and bandwidth and centre the picture on it. Use **Show known-frequency markers** and the group tick boxes above the audio picture to hide what you do not want. Labels that would overlap are skipped until you zoom in with a narrower Width of view.
6. **1 Hz step:** the **Step** list now goes down to 1 Hz for fine tuning.
7. **VFO lock:** press the big **VFO UNLOCKED / VFO LOCKED** button. When it is red and says LOCKED, the tuned frequency cannot change by clicking the picture, the mouse wheel, the Up/Down keys, typing, Jump to band or the memories. You can still move the view, change the mode and bandwidth, and record. The lock is remembered next time.
8. **Memory frequencies:** click one to tune it, set its mode and bandwidth and centre the picture on it. To add your own, tune a frequency, type an optional name and press **Save the tuned frequency to memory**. Press the × on a memory to delete it. A starter list (WWV, CHU, weather, airband and others) is included; delete what you do not want.

9. **Record what you hear:** press ● **Record audio** and later ■ **Stop recording**. The file is named with the date, time, frequency and mode, and appears in the **Audio recordings** box with **Play**, **Save a copy**, **Show in folder** and **Delete**. Files are kept in the **Recordings** folder in your RF Noise Map folder. They are small universal WAV files (8,000 samples a second, about 29 MB per hour), fine for voice and CW but not for high-fidelity music. Recording stops by itself if you close the panadapter or start logging.
10. To measure a frequency while driving, press **Mapping frequency 1–4**. The tuned frequency replaces that slot in the list in box 1. Pick a quiet one with no stations.

While you are logging the panadapter button changes to ▶ Watch a band while logging. Pick which of your frequencies to watch. You see the spectrum and waterfall (2 MHz around that frequency, which is marked by the red line), updated as the logger works, which is handy for spotting noise as you drive. You cannot tune or listen while logging, because the radio is busy measuring. Change the watched frequency at any time from the list.

> The level shown is **dBFS, not calibrated**. It is good for comparing places, not for absolute measurements. Recording **always uses fixed gain**, never automatic gain. The panadapter also uses the same fixed numbers by default, so what you see is what is recorded. Use **IF attenuation** (20–59 dB) and **RF attenuation** (0–27) to turn the radio down when strong stations overload it or the picture fills with false signals, and turn it back up to hear weak signals. These boxes are the same settings used for recording. **Automatic gain (for listening only)** is an optional extra that only affects the panadapter.

7. Where your data is stored

Everything you create is in one folder: on a Mac, the **RF Noise Map** folder in your home folder (not inside Documents: macOS blocks programs from Documents); on Windows, the install folder itself (normally **C:\RadioScripts\RFNoise**). The window's footer shows its exact path, and Open my data folder opens it.

Folder	What is in it
Drives	One **.csv** file per drive (opens in Excel / Numbers). Each row: time, latitude, longitude, speed, frequency, level. A small **.meta.json** beside it holds the name and notes.
Maps	Finished maps: one **.html** file each. Works by double-click in any browser.
Exports	Google Earth **.kml** files, and the **.zip** files made for emailing.
Backups	Zip backups made by the **Back up everything** button.
Settings	Your remembered choices and the logger's log file.

The program itself is kept apart from your data (Mac: Applications / RF Noise Map; Windows: the **app** folder inside RFNoise), so you can reinstall or update it without touching your data.

Level numbers: readings are in **dB relative to the receiver's maximum** (so they're negative; closer to zero = stronger). They compare places perfectly well. See section 10 for true dBm.

Your radios, and which radio measures which frequency

Box **1 Collect data** has a **Your radios** section. Tick what you have — one radio or both:

- **SDRplay RSP1** (USB): one receiver, so its frequencies take turns. Covers VHF, airband, 70 cm and the FM band.
- **Hermes Lite 2** (network cable): several receivers that run **at the same time**, so its frequencies are all read at the same instant with no retuning. HF only (about 0.1–38 MHz).

When both are ticked, the frequency table shows a **Radio** column, and each of the four slots picks its radio. **Auto** (the default) puts everything the Hermes Lite 2 can hear on the Hermes Lite 2 and the rest on the SDRplay. Or choose yourself — for example the Hermes Lite 2 on an AM broadcast channel and two HF ham bands, with the SDRplay on airband and 70 cm. A slot's radio cannot change in the middle of a drive, so the dots on each frequency always come from one radio and stay comparable.

1. Plug the radio's network cable into the computer (or its network adapter), connect the antenna and 12 V power, and wait about 20 seconds.
2. Close Thetis, SDR Console or any other program that uses the radio.
3. Tick **Hermes Lite 2**, press Find my radio. It says where it found it. If not, type the radio's address (for example 192.168.1.50) in the box and press it again.
4. **Detail per frequency**: 96 kHz is a good choice. Higher numbers use more of the network cable.
5. **Radio gain** (More settings): start at +15 dB. If the Slices view shows **OVERLOAD**, lower it. Leave it alone during a drive.
6. Press HL2_Test.bat (Windows, in the program folder) once to check the radio, the cable and the gain. Send its report if anything is wrong.

Slices. The **Slices** box shows one small panadapter for every frequency you are measuring, side by side, live. It appears automatically while logging, and Preview slices shows it before a drive so you can check the antenna and the gain. The shaded stripe is the part that is measured.

Hear and record a slice — even while driving. The Hermes Lite 2 streams all of its receivers all the time, so its slices have sound. Under each Hermes slice: Listen (one slice at a time, like tuning a radio), a mode choice (AM, SAM, USB, LSB, NFM — AM is right for power-line buzz), and ● Record. Recording keeps going while you drive and several slices can record at once, each to its own small WAV file in the Recordings folder, named with the slice, frequency and start time so it lines up with the map dots. Listening or recording never disturbs the measurements. Slices on the SDRplay have no sound: its single receiver is busy hopping between frequencies. The big listening panadapter below works with the SDRplay only, and only when not logging.

8. Make, view, print, save and email maps

Make a map

1. In box **2 Your recorded drives**, press Make map on a drive. The map opens in a new browser tab.
2. To **combine several drives** on one map, tick them and press Make map from ticked drives.
3. **How many dots?** Every reading is saved with its time to the millisecond, so the program can work out how far you travelled at every moment. Under the drives list, choose **One dot every [50] feet** to space the dots evenly *when the map is made*, whatever your speed was. Keep **the strongest reading** in each stretch (best for hunting noise: a short burst is never lost) or **the average**. The original recording is never changed, so you can make the same drive again with a different spacing.

Read the map

- **Blue** dots = weak, **red** dots = strong. The color bar shows the dB numbers; change **Min / Max** to stretch the colors, or press **Auto**.
- Pick the **frequency** in the drop-down. Each frequency has its own picture of the same route.
- Tap a dot to see its level, time and speed. **Hottest spots** lists the ten strongest places; tap one to zoom there.
- The button at top right switches **Street / Satellite**. Zoom in to separate dots.

Print or save as a PDF

In box **3 Maps** press Print / PDF (or the **Print / Save as PDF** button on the map itself). The print preview shows the title, color scale, strongest spots and the map. Choose your printer, or **Save as PDF** as the destination. Landscape paper works best.

Save a viewable file

Each map is already a file in **Maps** folder of your data folder (section 7). Save a copy downloads another copy anywhere you like. Double-click it any time to view it again. It needs internet only for the street background.

Email a map

1. Press Email beside the map. Type an address (optional) and keep **Put the map in a .zip** ticked (many email systems block .html attachments).
2. Press **Create email**. Your email program opens with the file attached; check it and press Send.
3. If your email program can't be controlled automatically (some web-based or newer mail apps), a blank email opens and the file is shown in a folder: drag it into the email. The person receiving it unzips it and opens the map in any browser.

Other extras

- **Google Earth:** Google Earth beside a drive creates a **.kml** file with the colored dots (in Exports). Open it with Google Earth.
- **Spreadsheet:** Save CSV gives the raw data for Excel / Numbers.
- **Name / notes:** add a name and notes (weather, antenna, what you were looking for).
- **Back up everything:** one zip of Drives, Maps and Exports in the Backups folder.
- **iPhone / iPad view:** tick **Let my iPhone/iPad open this window**, quit and reopen, then open the address shown on the iPhone (same Wi-Fi/hotspot). On Windows, allow the firewall question for private networks. Or simply email yourself the map.

9. Finding the noise source

1. **Drive a wide area** with a low band (such as 0.75 or 3.7 MHz) and one VHF channel. Low bands show the general neighborhood; VHF is more local.
2. **Make the map and look for the reddest area.** Use **Hottest spots** to jump to the top places.

3. **Drive that area again, slowly**, on every nearby street, and make a second map. The peak should narrow to a block or a pole.
4. **Check the car.** If readings stay high everywhere, the noise may be from the vehicle. Move the antenna, or test in a known-quiet place.
5. **Finish on foot** with your handheld and yagi. An **AM** mode works better than FM for tracking noise: an FM receiver is built to ignore the amplitude noise that AM shows. Aircraft-band AM (118–137 MHz) on a clear channel is a good choice.
6. Note the pole number or address and report it to the utility with your map (Print / PDF or Email).

10. Accuracy and calibration

- **Position:** a GPS is typically good to about 10–15 ft, so a dot can sit slightly off the road. Dots consistently 30 ft or more off the road suggest a GPS problem (poor sky view, or a slow-updating GPS in turns).
- **Spacing:** a new dot is saved each time you move the spacing distance (default 10 ft; **More settings**). Measuring four frequencies takes a fraction of a second, so at 45 mph spacing may be a little wider than 10 ft. Fewer frequencies = tighter dots.
- **Levels are relative** (dB below the receiver's maximum). To get true dBm: feed the receiver a known signal (a signal generator at, say, −60 dBm), note the reading, and enter **(known dBm) minus (reading)** in that frequency's **Calibration** box. Keep the same gain settings afterward.
- **Gain and attenuation:** automatic gain is always switched off while recording, so readings along the route can be compared. The **IF attenuation** and **RF attenuation** numbers (in **More settings**, and beside the panadapter) set how much the radio is turned down. If readings are stuck near the top or bottom of the scale, or strong stations overload the radio, raise the attenuation before the drive. You can change it while driving if you must, but dots before and after the change are then not directly comparable.

11. Troubleshooting

What you see	What to do
Mac: "cannot be opened because Apple cannot check it"	Right-click the file → **Open** → **Open**. Or use the Terminal method in section 2.
Windows: "Windows protected your PC"	**More info** → **Run anyway**.
Start logging says the radio library isn't available	The SDRplay driver isn't set up yet (section 3). Use **Practice mode** meanwhile. Press **Check setup**.
Radio library error mentioning "Python 3.9" or "DLL load failed"	Python and the radio library (PothosSDR) do not match. Run the installer again: it uses Python 3.9 and installs it if needed.
GPS: "Access is denied" or "port in use"	Another program (or an old RF Noise Map helper) has the GPS open. Quit and restart RF Noise Map, close other GPS programs, or re-plug the puck.
GPS: "nothing received" on a USB puck	Wrong speed or wrong port. Press **Find my GPS**.
"no SDRplay found"	Close other SDR programs (see the yellow box in the window). Plug it in directly (not through a weak hub). Make sure the SDRplay API is installed and the computer was restarted. Windows: the **SDRplay API Service** must be running.
GPS says "connecting" or "retrying" (iPhone)	Is the computer on the iPhone's hotspot? Is GPS2IP open and switched on, with Sockets on? Do the address and port match the app exactly?
GPS says "no GPS fix yet"	The GPS needs a view of the sky. Put it on the dashboard or roof; wait a minute or two.
Map shows dots but no streets	The street background needs internet when you open the map. Connect and reload. Use the layers button to try **Satellite**.
Dots are far from the road	See section 10 (GPS accuracy). Try the USB GPS, or place the GPS where it sees more sky.

Email button opens a blank email	Your email program can't be controlled automatically. Drag the file shown in the folder window into the email.
The window won't open / blank page	Windows: start it with **Start_RFNoise.bat** and read the messages in its window. Mac: look in `~/Library/Logs/RFNoiseMap.log`. Send it to Claude.

12. Updating, moving, uninstalling

- **Update:** run the new version's installer. It replaces the program and keeps your data.
- **Move to another computer:** install there, then copy the **Drives** folder over (or use a backup zip). Maps can be made on any computer, so a Windows PC can make and print maps from drives recorded on the Mac.
- **Uninstall:** run **Uninstall_RFNoise.bat** (Windows) or **Uninstall RF Noise Map** (Mac) from the install folder. It shows what is installed and how big your data is, then you choose: 1 remove the program only (your drives, maps and recordings are kept), 2 remove the program and your data (a backup zip is made first), 3 remove everything. On Windows it also removes the shortcuts, Start menu folder and the Windows Settings › Apps entry, and you can uninstall from there too. A record of the uninstall is kept.

Words used in this guide

dB / dBFS / dBm	Units of signal strength. dBFS is relative to the receiver's maximum (what this program records). dBm is an absolute power level (needs calibration).
CSV	A plain table file that Excel and Numbers open.
Hotspot (Personal Hotspot)	The iPhone sharing its internet over Wi-Fi.
KML	A file format Google Earth opens.
AM / FM	Radio modulation types. AM responds to amplitude noise (like power-line noise); FM mostly ignores it.